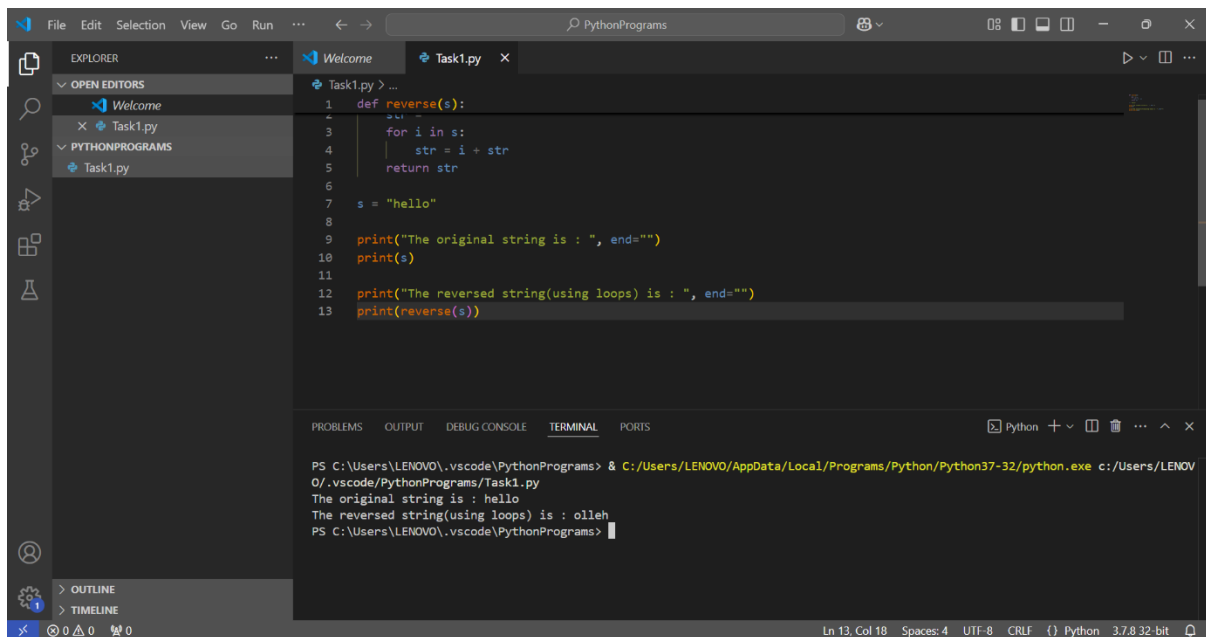


COGNIFY TECHNOLOGIES – PYTHON DEVELOPMENT INTERNSHIP

LEVEL 1

Task 1: String Reversal: Create a Python function that takes a string as input and returns the reverse of that string. For example, if the input is "hello," the function should return "olleh."



```
def reverse(s):
    str = ""
    for i in s:
        str = i + str
    return str

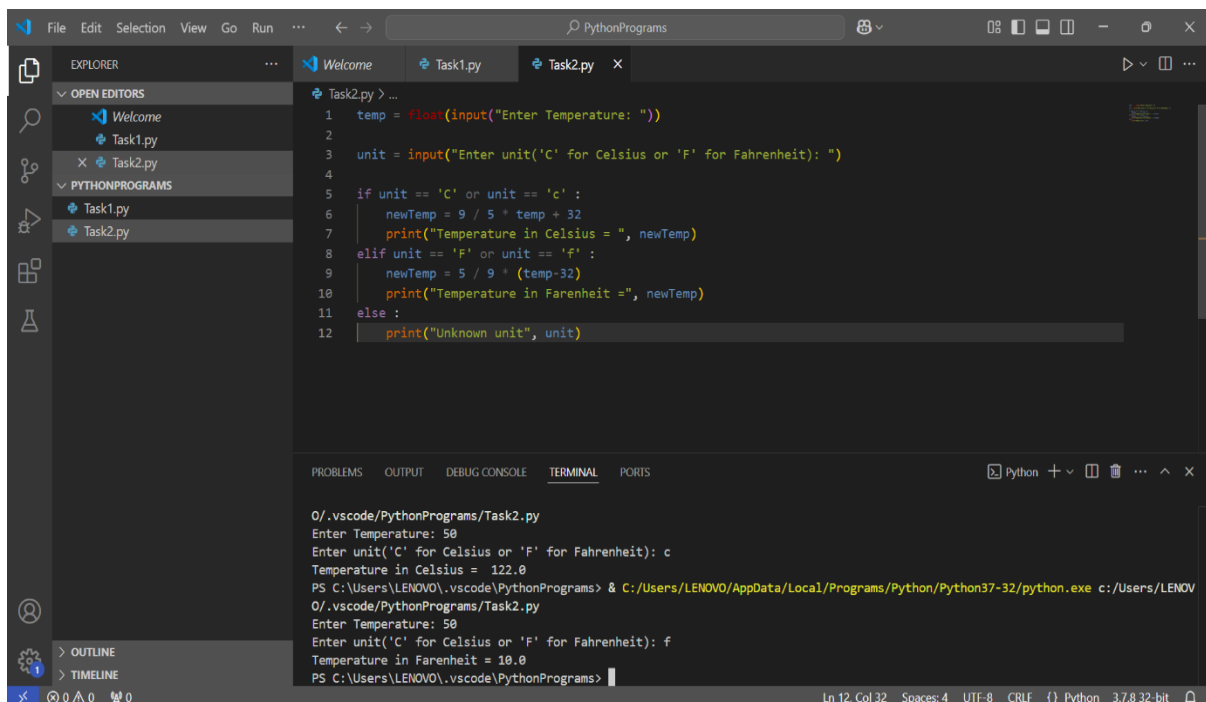
s = "hello"

print("The original string is : ", end="")
print(s)

print("The reversed string(using loops) is : ", end="")
print(reverse(s))
```

```
PS C:\Users\LENOVO\.vscode\PythonPrograms> & C:/Users/LENOVO/AppData/Local/Programs/Python/Python37-32/python.exe c:/Users/LENOVO/.vscode/PythonPrograms/Task1.py
The original string is : hello
The reversed string(using loops) is : olleh
PS C:\Users\LENOVO\.vscode\PythonPrograms>
```

Task 2: Temperature Conversion : Create a Python program that converts temperatures between Celsius and Fahrenheit. Prompt the user to enter a temperature value and the unit of measurement, and then display the converted temperature.



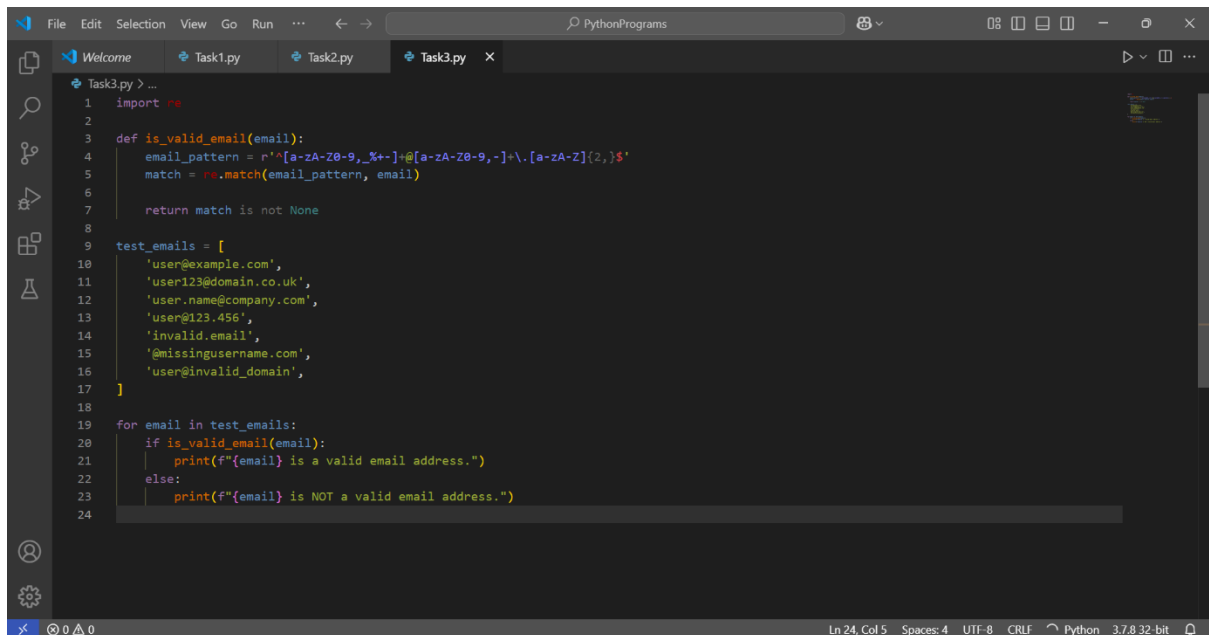
```
temp = float(input("Enter Temperature: "))
unit = input("Enter unit('C' for Celsius or 'F' for Fahrenheit): ")

if unit == 'C' or unit == 'c':
    newTemp = 9 / 5 * temp + 32
    print("Temperature in Celsius = ", newTemp)
elif unit == 'F' or unit == 'f':
    newTemp = 5 / 9 * (temp-32)
    print("Temperature in Farenheit =", newTemp)
else:
    print("Unknown unit", unit)
```

```
O:\.vscode\PythonPrograms\Task2.py
Enter Temperature: 50
Enter unit('C' for Celsius or 'F' for Fahrenheit): c
Temperature in Celsius = 122.0
PS C:\Users\LENOVO\.vscode\PythonPrograms> & C:/Users/LENOVO/AppData/Local/Programs/Python/Python37-32/python.exe c:/Users/LENOVO/.vscode/PythonPrograms/Task2.py
Enter Temperature: 50
Enter unit('C' for Celsius or 'F' for Fahrenheit): f
Temperature in Farenheit = 10.0
PS C:\Users\LENOVO\.vscode\PythonPrograms>
```

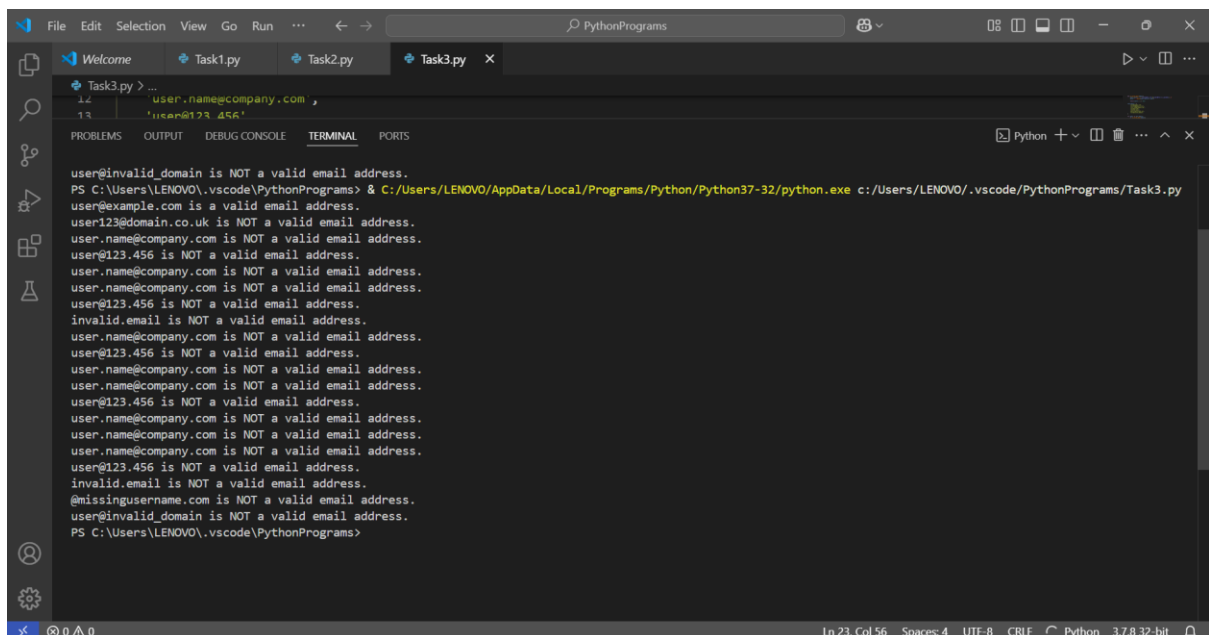
COGNIFY TECHNOLOGIES – PYTHON DEVELOPMENT INTERNSHIP

Task 3: Email Validator: Develop a Python function that validates whether a given string is a valid email address. Implement checks for the format, including the presence of an "@" symbol and a domain name.



```
1 import re
2
3 def is_valid_email(email):
4     email_pattern = r'^[a-zA-Z0-9_%.+]*@[a-zA-Z0-9-]+\.[a-zA-Z]{2,}$'
5     match = re.match(email_pattern, email)
6
7     return match is not None
8
9
10 test_emails = [
11     'user@example.com',
12     'user123@domain.co.uk',
13     'user.name@company.com',
14     'user@123.456',
15     'invalid.email',
16     '@missingusername.com',
17     'user@invalid_domain',
18 ]
19
20 for email in test_emails:
21     if is_valid_email(email):
22         print(f'{email} is a valid email address.')
23     else:
24         print(f'{email} is NOT a valid email address.')
```

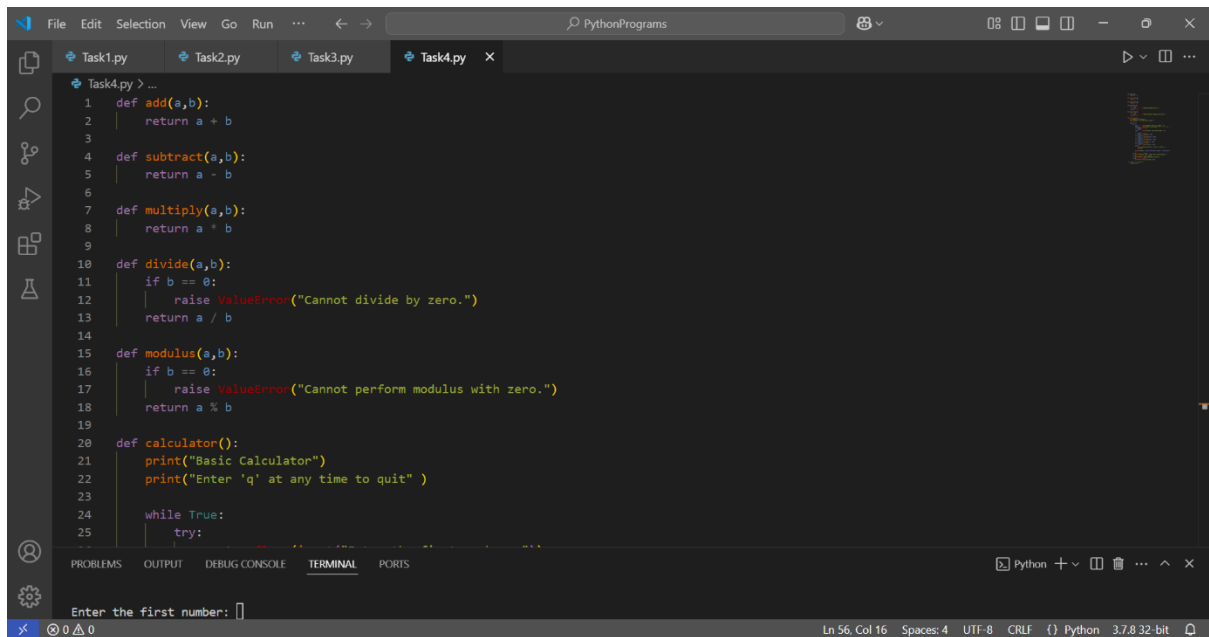
Output



```
user@invalid_domain is NOT a valid email address.
PS C:\Users\LENOVO\.vscode\PythonPrograms> & C:/Users/LENOVO/AppData/Local/Programs/Python/Python37-32/python.exe c:/Users/LENOVO/.vscode/PythonPrograms/Task3.py
user@example.com is a valid email address.
user123@domain.co.uk is NOT a valid email address.
user.name@company.com is NOT a valid email address.
user@123.456 is NOT a valid email address.
user.name@company.com is NOT a valid email address.
user.name@company.com is NOT a valid email address.
user@123.456 is NOT a valid email address.
invalid.email is NOT a valid email address.
user.name@company.com is NOT a valid email address.
user@123.456 is NOT a valid email address.
user.name@company.com is NOT a valid email address.
user.name@company.com is NOT a valid email address.
user@123.456 is NOT a valid email address.
user.name@company.com is NOT a valid email address.
user.name@company.com is NOT a valid email address.
user@123.456 is NOT a valid email address.
invalid.email is NOT a valid email address.
@missingusername.com is NOT a valid email address.
user@invalid_domain is NOT a valid email address.
PS C:\Users\LENOVO\.vscode\PythonPrograms>
```

COGNIFY TECHNOLOGIES – PYTHON DEVELOPMENT INTERNSHIP

Task 4: **Calculator Program:** Create a Python program that acts as a basic calculator. It should prompt the user to enter two numbers and an operator (+, -, *, /, %), and then display the result of the operation.

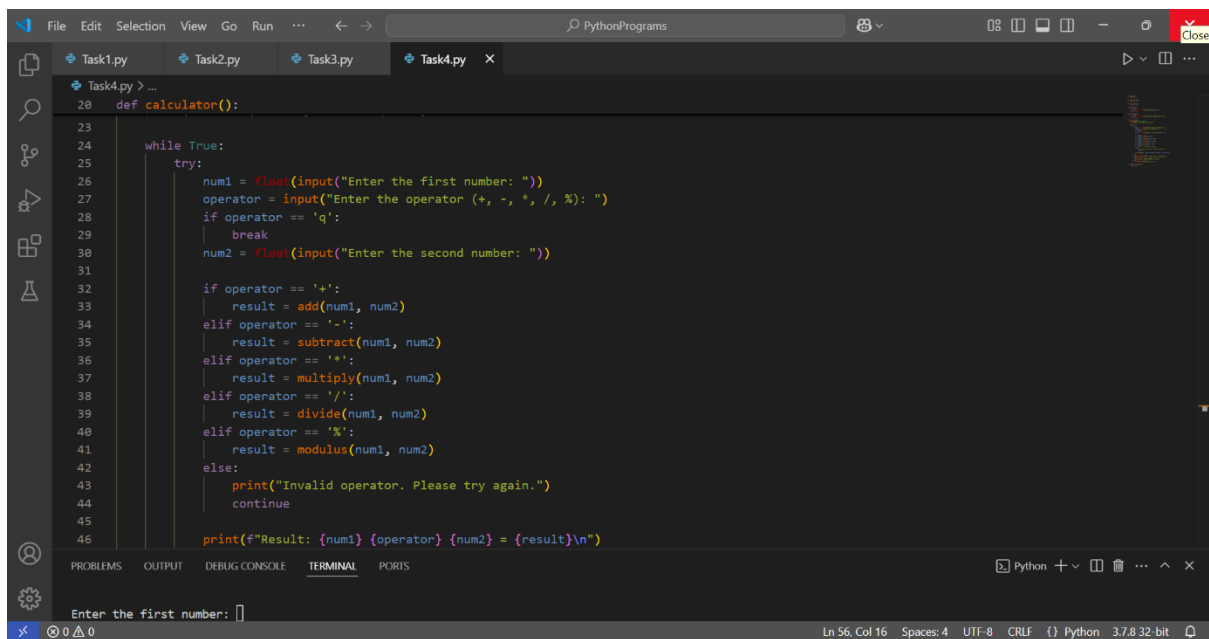


```
PythonPrograms
Task4.py x
Task4.py > ...
1 def add(a,b):
2     return a + b
3
4 def subtract(a,b):
5     return a - b
6
7 def multiply(a,b):
8     return a * b
9
10 def divide(a,b):
11     if b == 0:
12         raise ValueError("Cannot divide by zero.")
13     return a / b
14
15 def modulus(a,b):
16     if b == 0:
17         raise ValueError("Cannot perform modulus with zero.")
18     return a % b
19
20 def calculator():
21     print("Basic Calculator")
22     print("Enter 'q' at any time to quit" )
23
24     while True:
25         try:
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Enter the first number:

Ln 56, Col 16 Spaces: 4 UTF-8 CRLF {} Python 3.7.8 32-bit



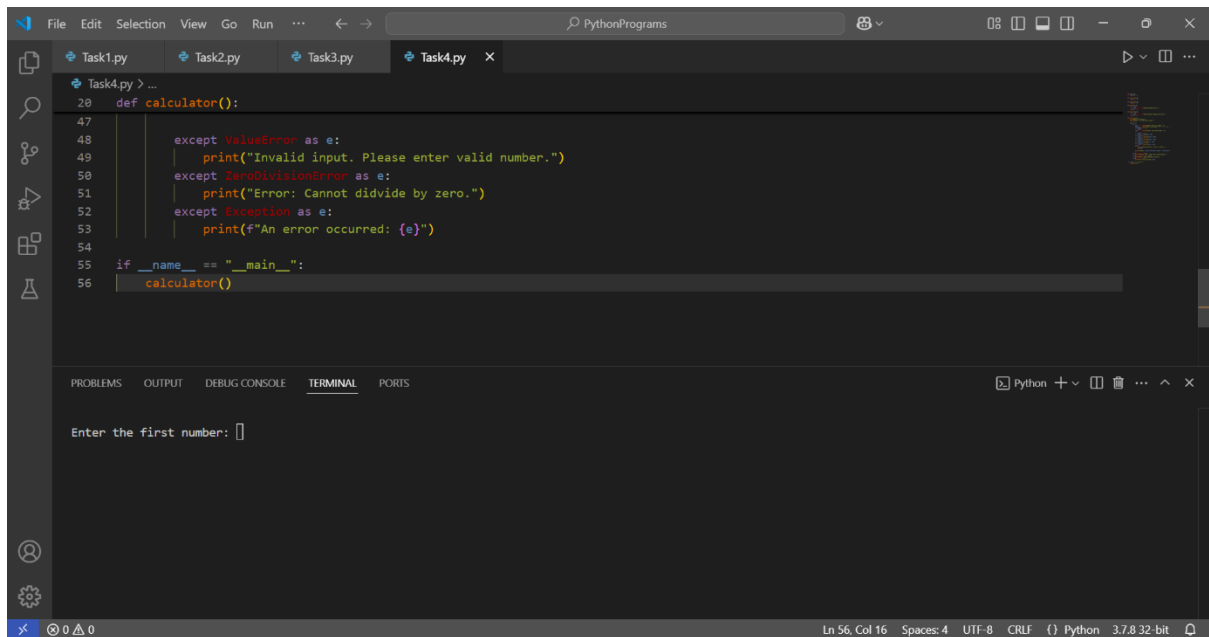
```
PythonPrograms
Task4.py x
Task4.py > ...
20 def calculator():
23
24     while True:
25         try:
26             num1 = float(input("Enter the first number: "))
27             operator = input("Enter the operator (+, -, *, /, %): ")
28             if operator == 'q':
29                 break
30             num2 = float(input("Enter the second number: "))
31
32             if operator == '+':
33                 result = add(num1, num2)
34             elif operator == '-':
35                 result = subtract(num1, num2)
36             elif operator == '*':
37                 result = multiply(num1, num2)
38             elif operator == '/':
39                 result = divide(num1, num2)
40             elif operator == '%':
41                 result = modulus(num1, num2)
42             else:
43                 print("Invalid operator. Please try again.")
44                 continue
45
46             print(f"Result: {num1} {operator} {num2} = {result}\n")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Enter the first number:

Ln 56, Col 16 Spaces: 4 UTF-8 CRLF {} Python 3.7.8 32-bit

COGNIFY TECHNOLOGIES – PYTHON DEVELOPMENT INTERNSHIP



The screenshot shows a Visual Studio Code editor window with the file 'Task4.py' open. The code is a Python calculator function with error handling for ValueError, ZeroDivisionError, and a general Exception. The terminal at the bottom shows the prompt 'Enter the first number: '.

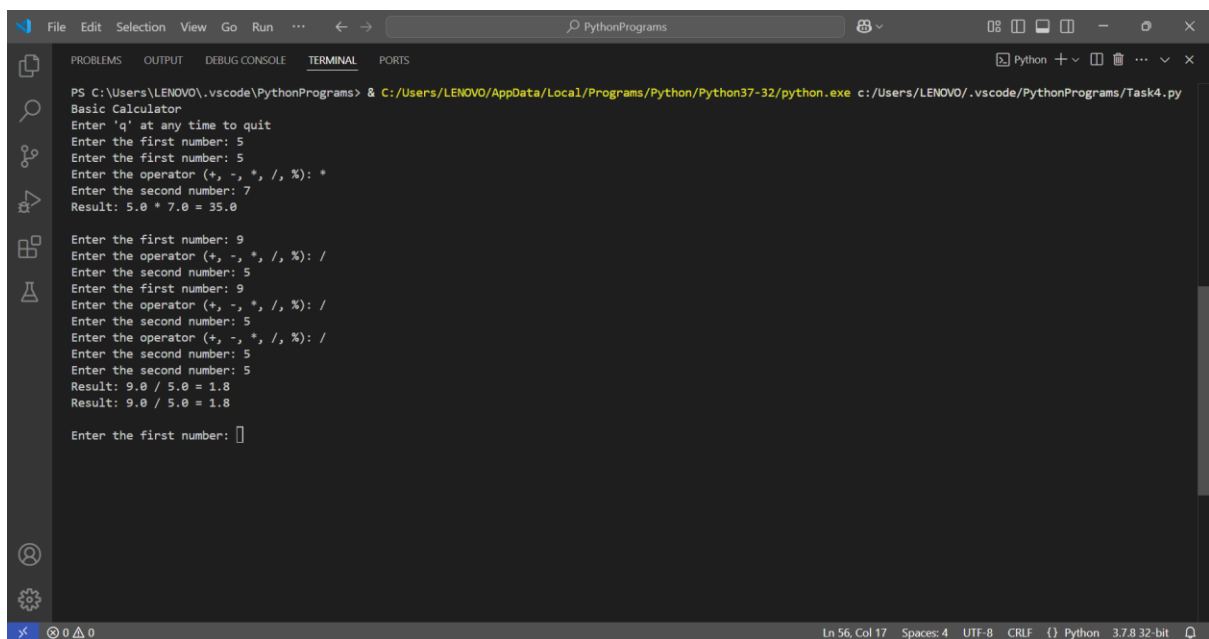
```
20 def calculator():
47
48     except ValueError as e:
49         print("Invalid input. Please enter valid number.")
50     except ZeroDivisionError as e:
51         print("Error: Cannot divide by zero.")
52     except Exception as e:
53         print(f"An error occurred: {e}")
54
55 if __name__ == "__main__":
56     calculator()
```

PROBLEMS OUTPUT DEBUG CONSOLE **TERMINAL** PORTS

Enter the first number:

Ln 56, Col 16 Spaces: 4 UTF-8 CRLF {} Python 3.7.8 32-bit

Output



The screenshot shows the terminal output of the calculator program. It prompts the user for the first number, operator, and second number, and then displays the result. The output shows three successful calculations: 5 * 7 = 35.0, 9 / 5 = 1.8, and 9.0 / 5.0 = 1.8.

```
PS C:\Users\LENOVO\.vscode\PythonPrograms> & C:/Users/LENOVO/AppData/Local/Programs/Python/Python37-32/python.exe c:/Users/LENOVO/.vscode/PythonPrograms/Task4.py
Basic Calculator
Enter 'q' at any time to quit
Enter the first number: 5
Enter the first number: 5
Enter the operator (+, -, *, /, %): *
Enter the second number: 7
Result: 5.0 * 7.0 = 35.0

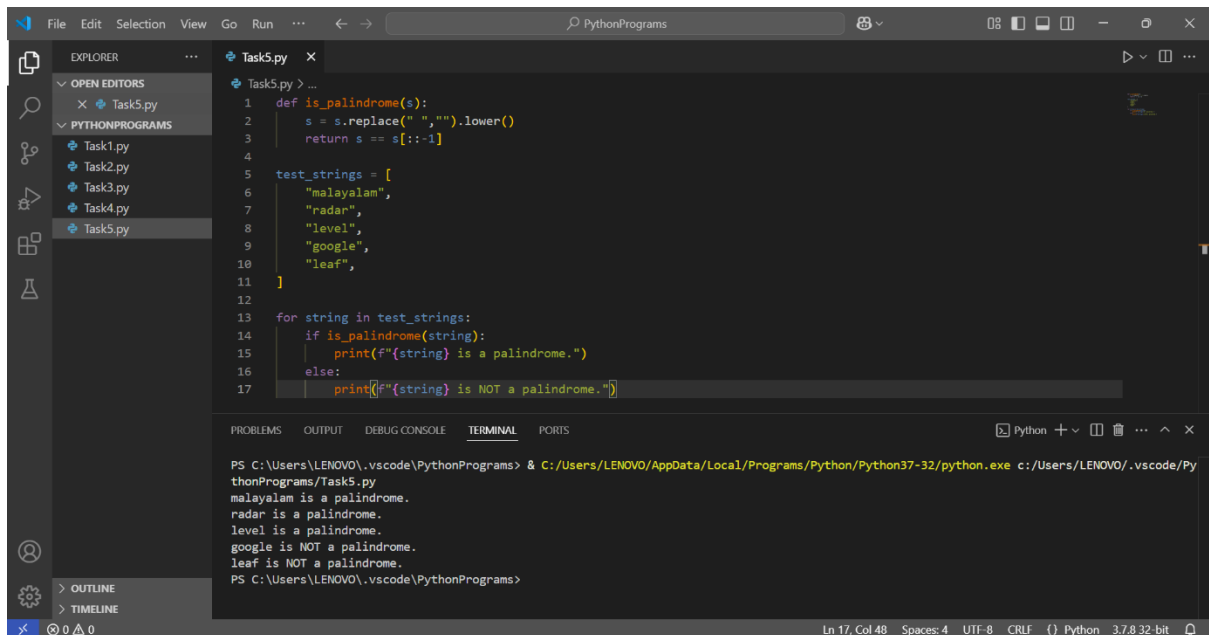
Enter the first number: 9
Enter the operator (+, -, *, /, %): /
Enter the second number: 5
Enter the first number: 9
Enter the operator (+, -, *, /, %): /
Enter the second number: 5
Enter the operator (+, -, *, /, %): /
Enter the second number: 5
Result: 9.0 / 5.0 = 1.8
Result: 9.0 / 5.0 = 1.8

Enter the first number: 
```

Ln 56, Col 17 Spaces: 4 UTF-8 CRLF {} Python 3.7.8 32-bit

COGNIFY TECHNOLOGIES – PYTHON DEVELOPMENT INTERNSHIP

Task 5: Palindrome Checker : Write a Python function that checks whether a given string is a palindrome. A palindrome is a word, phrase, or sequence that reads the same backward as forward (e.g., "madam" or "racecar").



```
1 def is_palindrome(s):
2     s = s.replace(" ", "").lower()
3     return s == s[::-1]
4
5 test_strings = [
6     "malayalam",
7     "radar",
8     "level",
9     "google",
10    "leaf",
11 ]
12
13 for string in test_strings:
14     if is_palindrome(string):
15         print(f"{string} is a palindrome.")
16     else:
17         print(f"{string} is NOT a palindrome.")
```

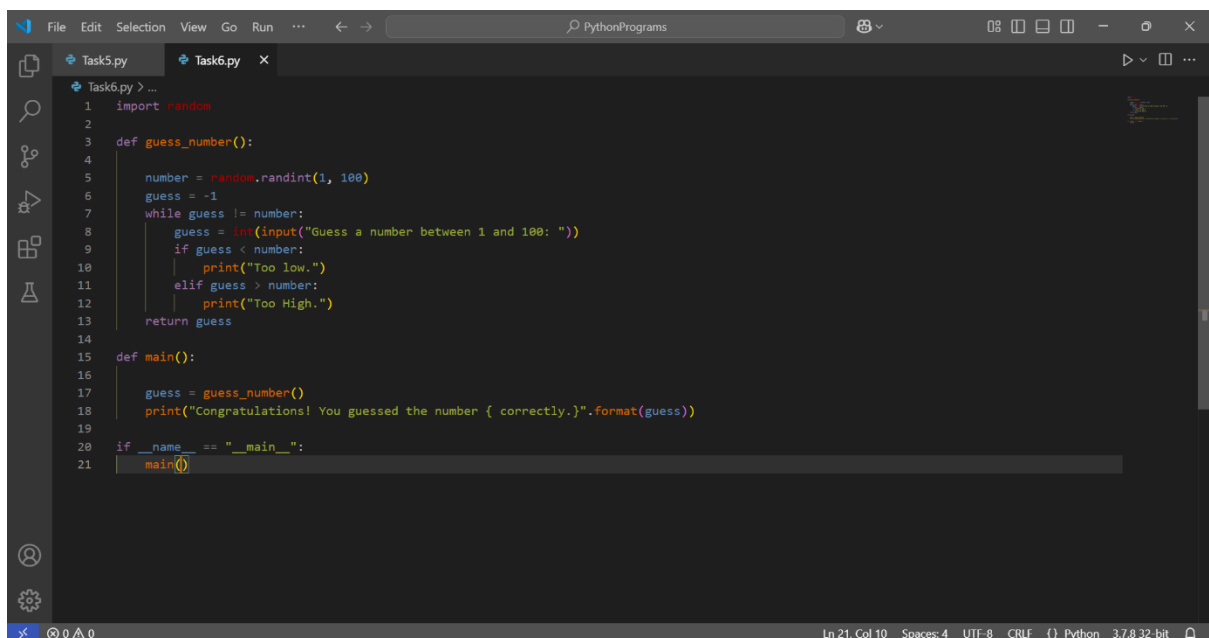
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\LENOVO\.vscode\PythonPrograms> & C:/Users/LENOVO/AppData/Local/Programs/Python/Python37-32/python.exe c:/Users/LENOVO/.vscode/PythonPrograms/Task5.py
malayalam is a palindrome.
radar is a palindrome.
level is a palindrome.
google is NOT a palindrome.
leaf is NOT a palindrome.
PS C:\Users\LENOVO\.vscode\PythonPrograms>
```

Ln 17, Col 48 Spaces: 4 UTF-8 CRLF {} Python 3.7.8 32-bit

LEVEL 2

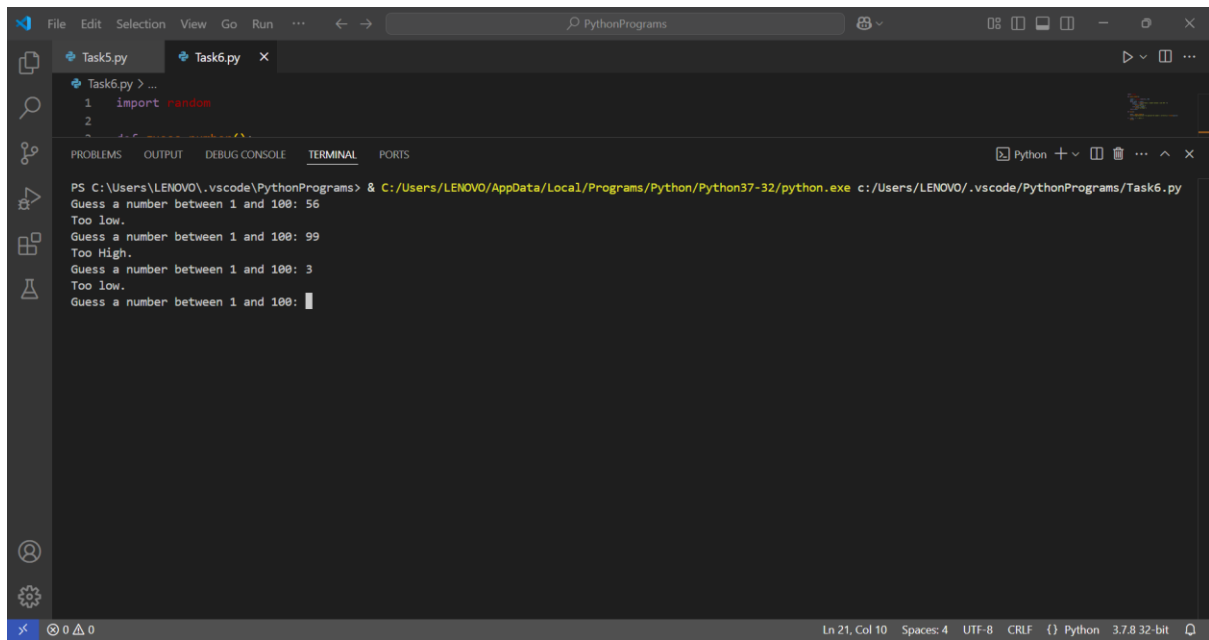
Task1:Guessing Game : Write a Python program that generates a random number between 1 and 100. The user should then try to guess the number. The program should provide hints such as "too high" or "too low" until the correct number is guessed.



```
1 import random
2
3 def guess_number():
4
5     number = random.randint(1, 100)
6     guess = -1
7     while guess != number:
8         guess = int(input("Guess a number between 1 and 100: "))
9         if guess < number:
10             print("Too low.")
11         elif guess > number:
12             print("Too High.")
13     return guess
14
15 def main():
16
17     guess = guess_number()
18     print("Congratulations! You guessed the number { correctly.}".format(guess))
19
20 if __name__ == "__main__":
21     main()
```

Ln 21, Col 10 Spaces: 4 UTF-8 CRLF {} Python 3.7.8 32-bit

COGNIFY TECHNOLOGIES – PYTHON DEVELOPMENT INTERNSHIP



The screenshot displays the Visual Studio Code (VS Code) interface. The top menu bar includes File, Edit, Selection, View, Go, Run, and other standard options. The Explorer sidebar on the left shows two files: Task5.py and Task6.py. The main editor area displays the code for Task6.py, which includes an import statement for the random module. Below the editor, the TERMINAL panel is active, showing the command prompt output of the program. The program prompts the user to guess a number between 1 and 100, and the terminal shows three attempts: 56 (Too low), 99 (Too High), and 3 (Too low). The status bar at the bottom indicates the current line and column (Ln 21, Col 10), encoding (UTF-8), line endings (CRLF), and the Python version (3.7.8 32-bit).

```
File Edit Selection View Go Run ... PythonPrograms
```

Task5.py Task6.py

```
1 import random
2
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python

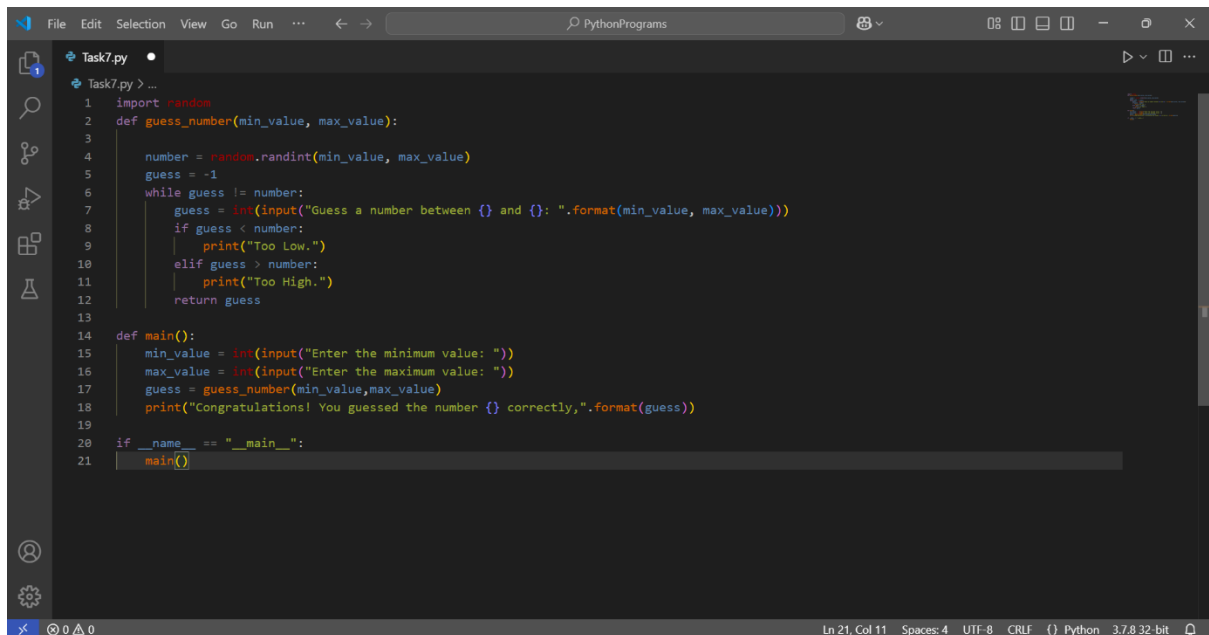
PS C:\Users\LENOVO\.vscode\PythonPrograms> & C:/Users/LENOVO/AppData/Local/Programs/Python/Python37-32/python.exe c:/Users/LENOVO/.vscode/PythonPrograms/Task6.py

Guess a number between 1 and 100: 56
Too low.
Guess a number between 1 and 100: 99
Too High.
Guess a number between 1 and 100: 3
Too low.
Guess a number between 1 and 100:

Ln 21, Col 10 Spaces: 4 UTF-8 CRLF {} Python 3.7.8 32-bit

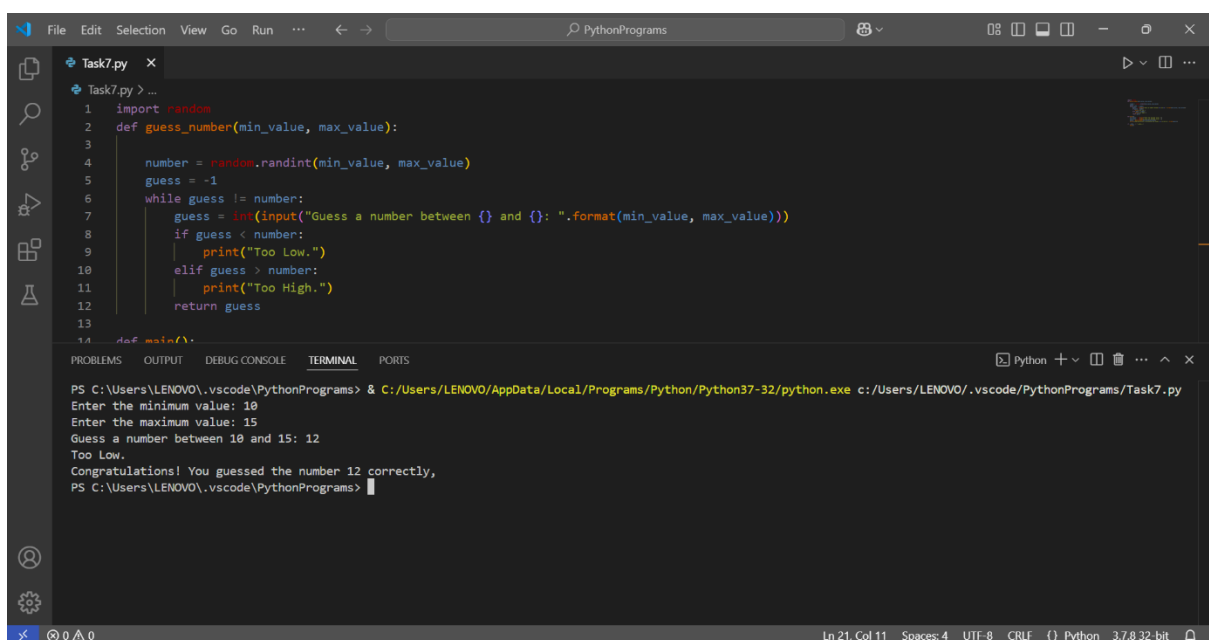
COGNIFY TECHNOLOGIES – PYTHON DEVELOPMENT INTERNSHIP

Task 2: **Number Guesser** : Create a number guessing game where the program generates a random number between a specified range, and the user tries to guess it. Provide feedback to the user if their guess is too high or too low.



```
1 import random
2 def guess_number(min_value, max_value):
3
4     number = random.randint(min_value, max_value)
5     guess = -1
6     while guess != number:
7         guess = int(input("Guess a number between {} and {}: ".format(min_value, max_value)))
8         if guess < number:
9             print("Too Low.")
10        elif guess > number:
11            print("Too High.")
12        return guess
13
14 def main():
15     min_value = int(input("Enter the minimum value: "))
16     max_value = int(input("Enter the maximum value: "))
17     guess = guess_number(min_value,max_value)
18     print("Congratulations! You guessed the number {} correctly.".format(guess))
19
20 if __name__ == "__main__":
21     main()
```

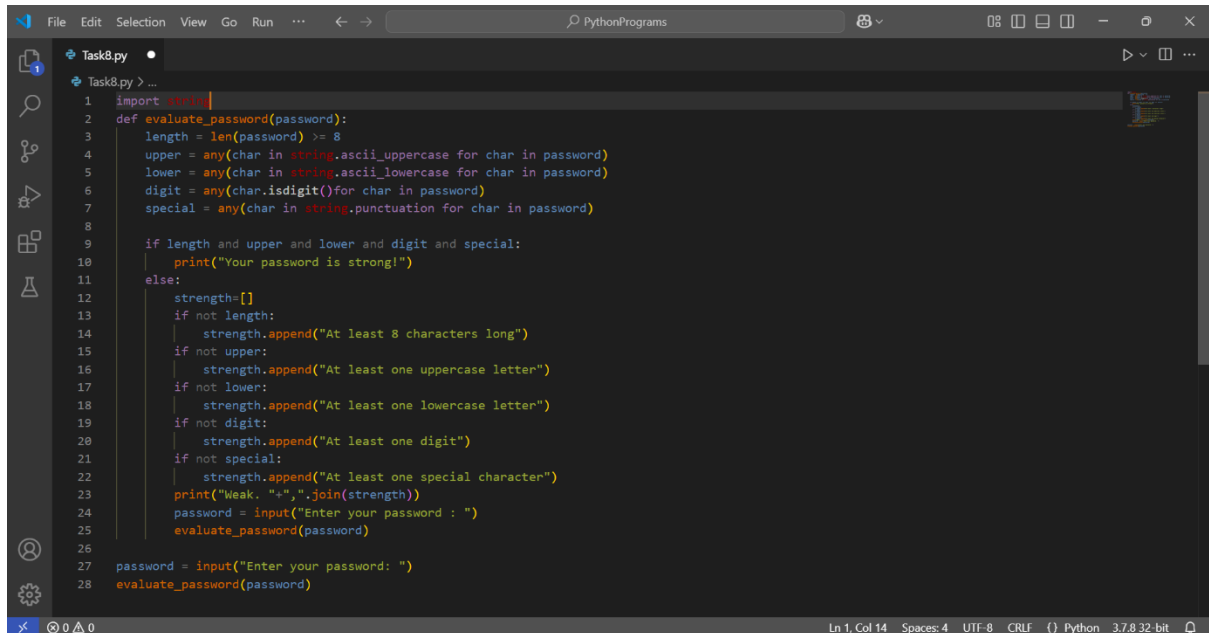
Output



```
PS C:\Users\LENOVO\.vscode\PythonPrograms> & C:/Users/LENOVO/AppData/Local/Programs/Python/Python37-32/python.exe c:/Users/LENOVO/.vscode/PythonPrograms/Task7.py
Enter the minimum value: 10
Enter the maximum value: 15
Guess a number between 10 and 15: 12
Too Low.
Congratulations! You guessed the number 12 correctly,
PS C:\Users\LENOVO\.vscode\PythonPrograms>
```

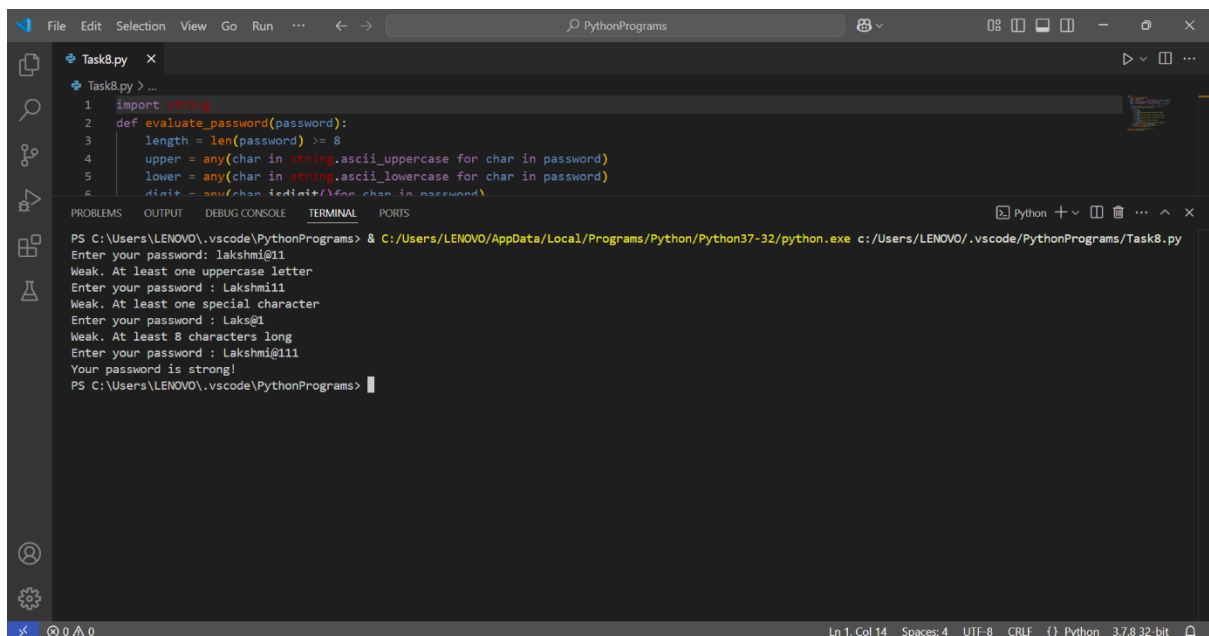
COGNIFY TECHNOLOGIES – PYTHON DEVELOPMENT INTERNSHIP

Task 3: Password Strength: Checker Create a Python function that evaluates the strength of a password entered by the user. Implement checks for factors such as length, presence of uppercase and lowercase letters, digits, and special characters.



```
1 import string
2 def evaluate_password(password):
3     length = len(password) >= 8
4     upper = any(char in string.ascii_uppercase for char in password)
5     lower = any(char in string.ascii_lowercase for char in password)
6     digit = any(char.isdigit() for char in password)
7     special = any(char in string.punctuation for char in password)
8
9     if length and upper and lower and digit and special:
10         print("Your password is strong!")
11     else:
12         strength=[]
13         if not length:
14             strength.append("At least 8 characters long")
15         if not upper:
16             strength.append("At least one uppercase letter")
17         if not lower:
18             strength.append("At least one lowercase letter")
19         if not digit:
20             strength.append("At least one digit")
21         if not special:
22             strength.append("At least one special character")
23         print("Weak. "+",".join(strength))
24         password = input("Enter your password : ")
25         evaluate_password(password)
26
27 password = input("Enter your password: ")
28 evaluate_password(password)
```

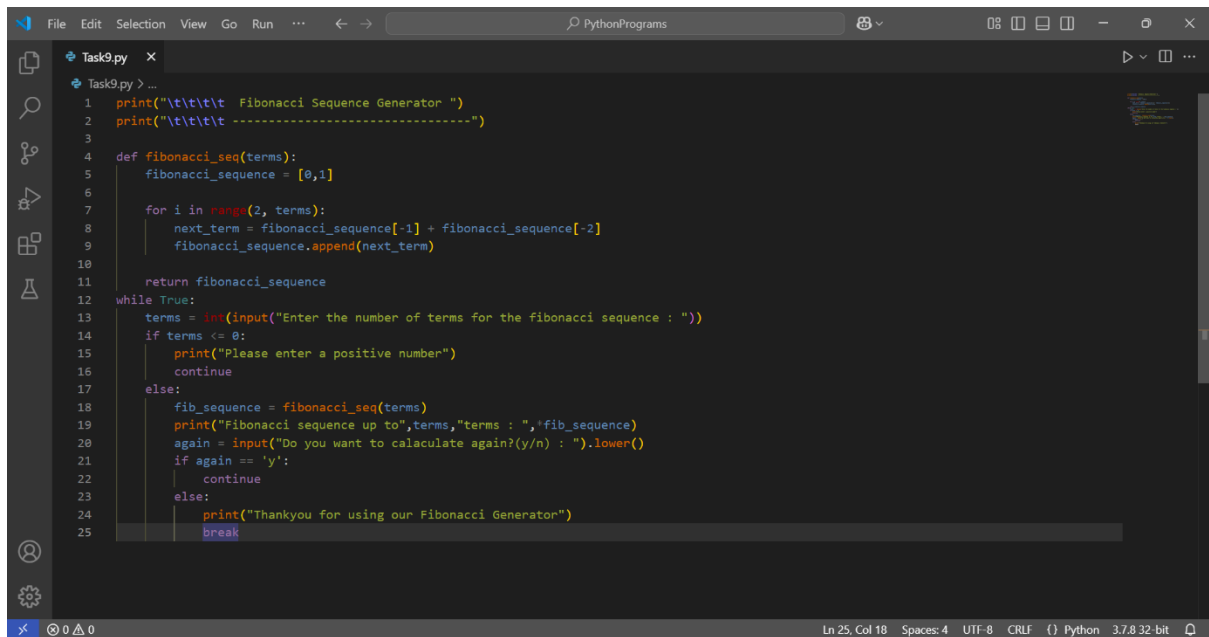
Output



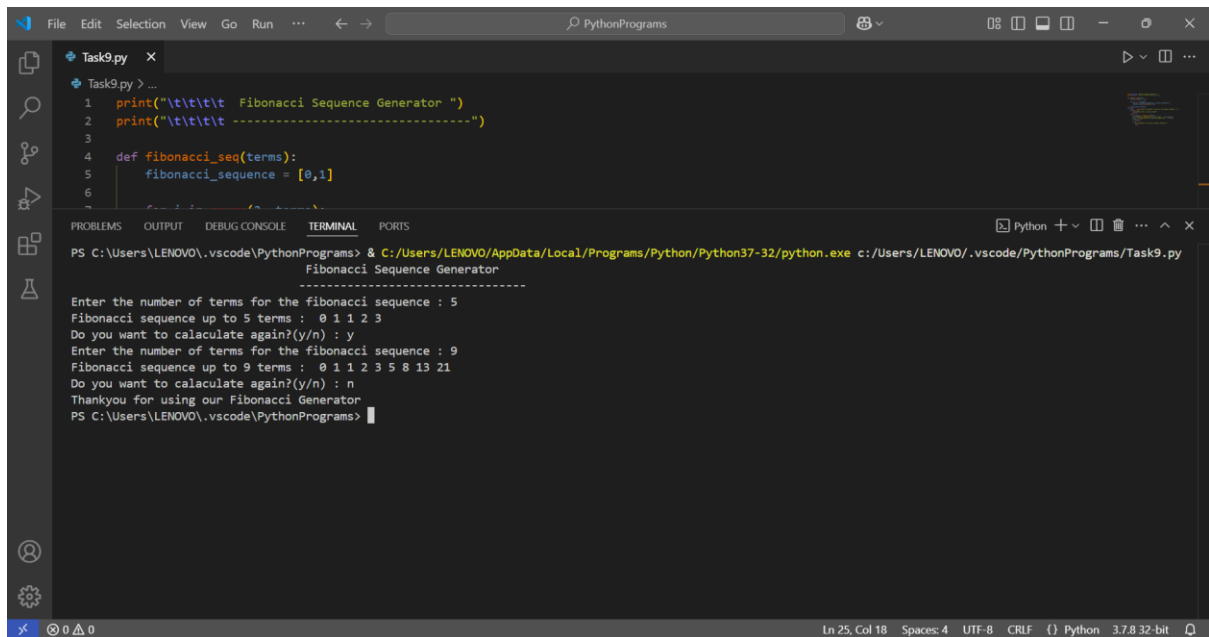
```
PS C:\Users\LENOVO\.vscode\PythonPrograms> & C:/Users/LENOVO/AppData/Local/Programs/Python/Python37-32/python.exe c:/Users/LENOVO/.vscode/PythonPrograms/Task8.py
Enter your password: lakshmi@11
Weak. At least one uppercase letter
Enter your password : Lakshmi11
Weak. At least one special character
Enter your password : Laks@1
Weak. At least 8 characters long
Enter your password : Lakshmi@111
Your password is strong!
PS C:\Users\LENOVO\.vscode\PythonPrograms>
```


COGNIFY TECHNOLOGIES – PYTHON DEVELOPMENT INTERNSHIP

Task 4: Fibonacci Sequence: Write a Python function that generates the Fibonacci sequence up to a given number of terms. The function should take an integer input from the user and display the Fibonacci sequence up to that number of terms.



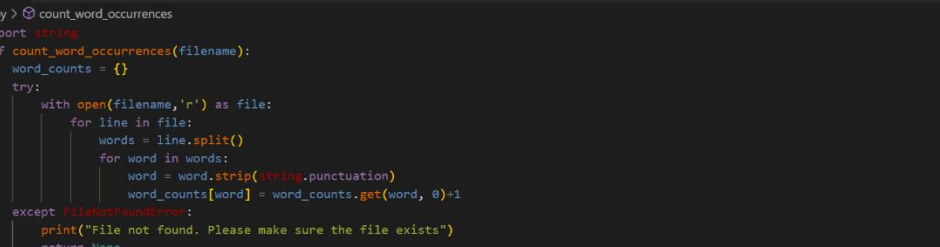
```
1 print("\t\t\t\t\t Fibonacci Sequence Generator ")
2 print("\t\t\t\t\t -----")
3
4 def fibonacci_seq(terms):
5     fibonacci_sequence = [0,1]
6
7     for i in range(2, terms):
8         next_term = fibonacci_sequence[-1] + fibonacci_sequence[-2]
9         fibonacci_sequence.append(next_term)
10
11     return fibonacci_sequence
12
13 while True:
14     terms = int(input("Enter the number of terms for the fibonacci sequence : "))
15     if terms <= 0:
16         print("Please enter a positive number")
17         continue
18     else:
19         fib_sequence = fibonacci_seq(terms)
20         print("Fibonacci sequence up to",terms,"terms : ",fib_sequence)
21         again = input("Do you want to calaculate again?(y/n) : ").lower()
22         if again == 'y':
23             continue
24         else:
25             print("Thankyou for using our Fibonacci Generator")
26             break
```



```
PS C:\Users\LENOVO\.vscode\PythonPrograms> & C:/Users/LENOVO/AppData/Local/Programs/Python/Python37-32/python.exe c:/Users/LENOVO/.vscode/PythonPrograms/Task9.py
Fibonacci Sequence Generator
-----
Enter the number of terms for the fibonacci sequence : 5
Fibonacci sequence up to 5 terms : 0 1 1 2 3
Do you want to calaculate again?(y/n) : y
Enter the number of terms for the fibonacci sequence : 9
Fibonacci sequence up to 9 terms : 0 1 1 2 3 5 8 13 21
Do you want to calaculate again?(y/n) : n
Thankyou for using our Fibonacci Generator
PS C:\Users\LENOVO\.vscode\PythonPrograms>
```

COGNIFY TECHNOLOGIES – PYTHON DEVELOPMENT INTERNSHIP

Task 5: File Manipulation Write a Python program that reads a text file and counts the occurrences of each word in the file. Display the results in alphabetical order along with their respective counts.

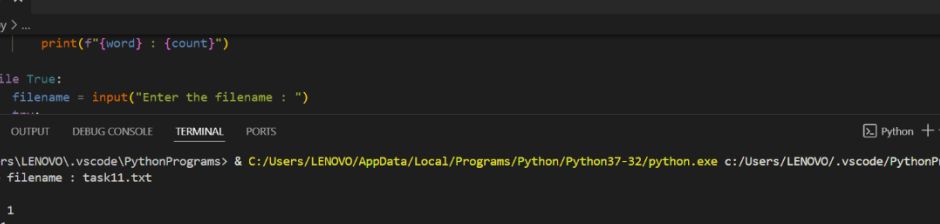


The screenshot shows a Python IDE with a menu bar (File, Edit, Selection, View, Go, Run, ...) and a toolbar. The main editor window displays a Python script named `Task10.py` with the following code:

```
1 import string
2 def count_word_occurrences(filename):
3     word_counts = {}
4     try:
5         with open(filename, 'r') as file:
6             for line in file:
7                 words = line.split()
8                 for word in words:
9                     word = word.strip(string.punctuation)
10                    word_counts[word] = word_counts.get(word, 0) + 1
11 except FileNotFoundError:
12     print("File not found. Please make sure the file exists")
13     return None
14 return word_counts
15
16 def display_word_count(word_counts):
17     sorted_word_counts = sorted(word_counts.items(), key=lambda x: x[0].lower())
18     for word, count in sorted_word_counts:
19         print(f"{word} : {count}")
20
21 while True:
22     filename = input("Enter the filename : ")
23     try:
24         word_count = count_word_occurrences(filename)
25         display_word_count(word_count)
26         break
27     except FileNotFoundError:
28         print("File not found")
29         choice = input("Do you want to enter another filename?(y/n) : ").lower()
30         if choice != 'y':
31             break
```

The status bar at the bottom indicates the current cursor position is Line 14, Column 23, with 4 spaces, UTF-8 encoding, CRLF line endings, Python 3.7.8, and 32-bit architecture.

Output



The screenshot shows the VS Code interface with a Python file named `Task10.py` open. The code in the editor is as follows:

```
19     print(f"{word} : {count}")
20
21 while True:
22     filename = input("Enter the filename : ")
23     ...
```

The **TERMINAL** panel at the bottom shows the command prompt output:

```
PS C:\Users\LENOVO\AppData\Local\Programs\Python\Python37-32\python.exe c:/Users/LENOVO/.vscode/PythonPrograms/Task10.py
Enter the filename : task11.txt
a : 1
Awesome : 1
Coding : 1
day : 2
every : 1
Have : 1
if : 1
is : 1
Make : 1
nice : 1
rest : 1
rust : 1
Welcome : 1
worthful : 1
you : 2
PS C:\Users\LENOVO\.vscode\PythonPrograms>
```